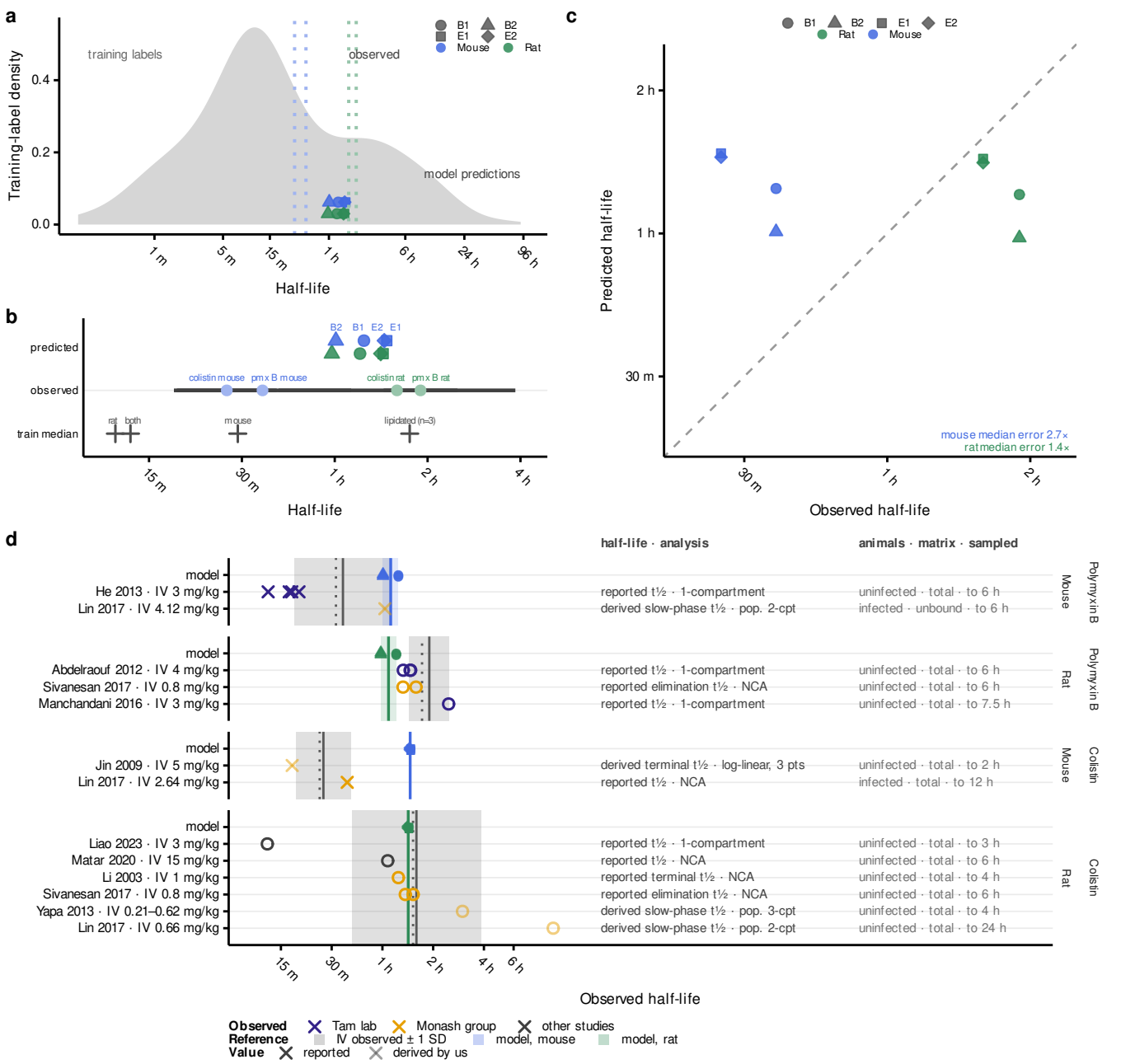
a
training labels
observed
model predictions
B1
B2
E1
E2
Mouse
Rat
Training-label density
0.0
0.2
0.4
1 m
5 m
15 m
1 h
6 h
24 h
96 h
Half-life
b
predicted
observed
train median
B2 B1 E2 E1
colistin mouse
pm x B mouse
colistin rat
pm x B rat
rat both
mouse
lipidated (n=3)
15 m
30 m
1 h
2 h
4 h
Half-life
c
B1
B2
E1
E2
Rat
Mouse
Predicted half-life
2 h
1 h
30 m
mouse median error 2.7×
rat median error 1.4×
30 m
1 h
2 h
Observed half-life
d
half-life · analysis
animals · matrix · sampled
model
He 2013 · IV 3 mg/kg
Lin 2017 · IV 4.12 mg/kg
reported t½ · 1-compartment
derived slow-phase t½ · pop. 2-cpt
uninfected · total · to 6 h
infected · unbound · to 6 h
Mouse
Polymyxin B
model
Abdelraouf 2012 · IV 4 mg/kg
Sivanesan 2017 · IV 0.8 mg/kg
Manchandani 2016 · IV 3 mg/kg
reported t½ · 1-compartment
reported elimination t½ · NCA
reported t½ · 1-compartment
uninfected · total · to 6 h
uninfected · total · to 6 h
uninfected · total · to 7.5 h
Rat
Polymyxin B
model
Jin 2009 · IV 5 mg/kg
Lin 2017 · IV 2.64 mg/kg
derived terminal t½ · log-linear, 3 pts
reported t½ · NCA
uninfected · total · to 2 h
infected · total · to 12 h
Mouse
Colistin
model
Liao 2023 · IV 3 mg/kg
Matar 2020 · IV 15 mg/kg
Li 2003 · IV 1 mg/kg
Sivanesan 2017 · IV 0.8 mg/kg
Yapa 2013 · IV 0.21–0.62 mg/kg
Lin 2017 · IV 0.66 mg/kg
reported t½ · 1-compartment
reported t½ · NCA
reported terminal t½ · NCA
reported elimination t½ · NCA
derived slow-phase t½ · pop. 3-cpt
derived slow-phase t½ · pop. 2-cpt
uninfected · total · to 3 h
uninfected · total · to 6 h
uninfected · total · to 4 h
uninfected · total · to 6 h
uninfected · total · to 4 h
uninfected · total · to 24 h
Rat
Colistin
15 m
30 m
1 h
2 h
4 h
6 h
Observed half-life
Observed
Tam lab
Monash group
other studies
Reference
IV observed ± 1 SD
model, mouse
model, rat
Value
reported
derived by us